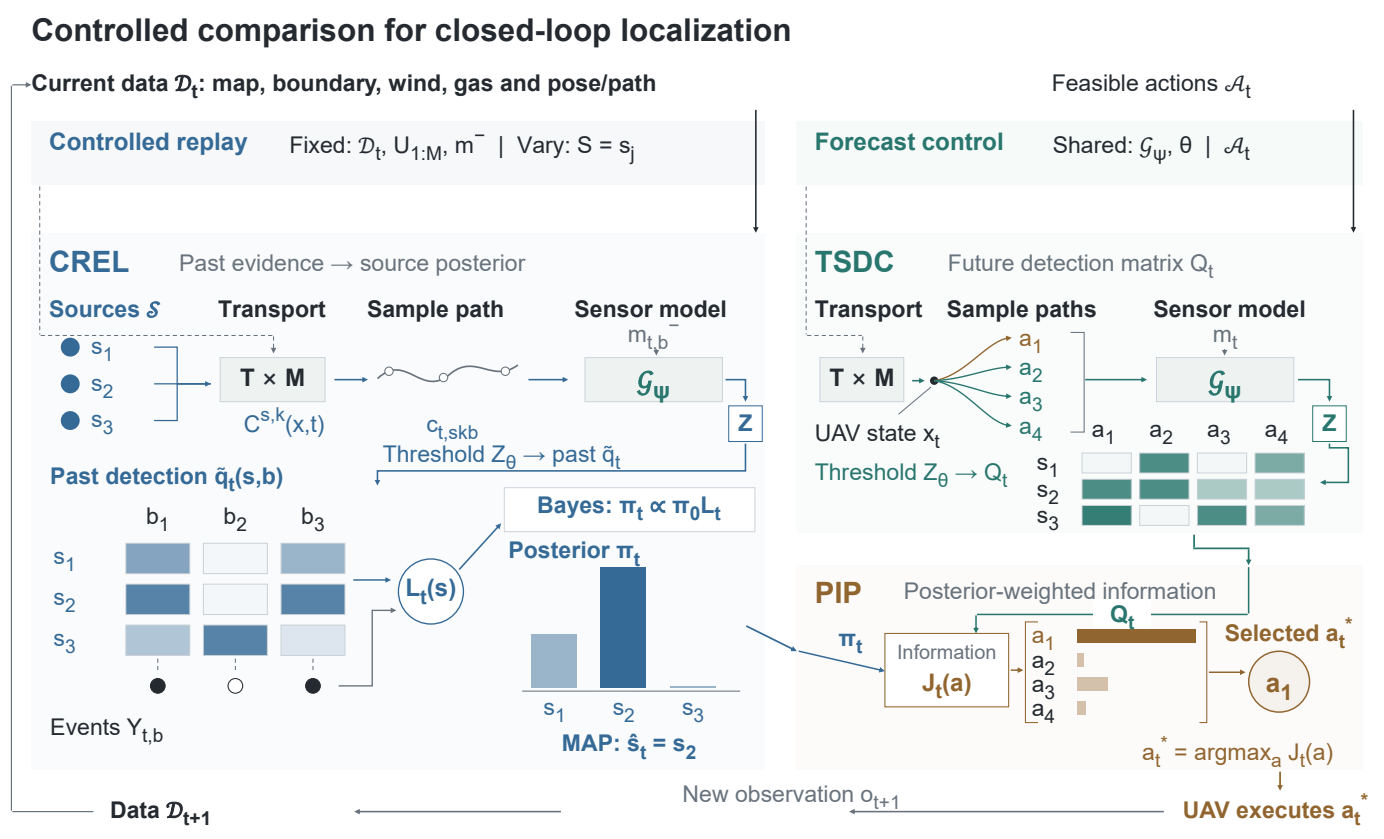

Controlled comparison for closed-loop localization
Current data $\mathcal{D}_t$: map, boundary, wind, gas and pose/path
Feasible actions $\mathcal{A}_t$
Controlled replay
Fixed: $\mathcal{D}_t$, $U_{1:M}$, m^- | Vary: $S = s_j$
Forecast control
Shared: $\mathcal{G}_\psi$, θ | $\mathcal{A}_t$
CREL
Past evidence → source posterior
Sources $\mathcal{S}$
Transport
Sample path
Sensor model
s_1
s_2
s_3
T × M
$C^{s,k}(x,t)$
$\mathcal{G}_\psi$
$m_{t,b}^-$
Z
$c_{t,skb}$
Threshold Z_θ → past $\tilde{q}_t$
Past detection $\tilde{q}_t(s,b)$
b_1
b_2
b_3
$L_t(s)$
Bayes: $\pi_t \propto \pi_0 L_t$
Posterior π_t
s_1
s_2
s_3
MAP: $\hat{s}_t = s_2$
Events $Y_{t,b}$
TSDC
Future detection matrix Q_t
Transport
Sample paths
Sensor model
T × M
UAV state x_t
a_1
a_2
a_3
a_4
m_t
$\mathcal{G}_\psi$
Z
Threshold Z_θ → Q_t
PIP
Posterior-weighted information
Q_t
π_t
Information
$J_t(a)$
Selected a_t^*
$a_t^* = \text{argmax}_a J_t(a)$
UAV executes a_t^*
New observation o_{t+1}
Data $\mathcal{D}_{t+1}$